ALBUKHARY INTERNATIONAL UNIVERSITY

SCHOOL OF COMPUTING AND INFORMATICS

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1. Introduction

Data mining is the process of discovering meaningful patterns, relationships, and knowledge from large datasets to support decision-making and predictive analysis. With the increasing availability of digital data, data mining techniques have become essential across various domains, including business, finance, healthcare, education, and public policy. Among the most widely used applications of data mining is classification, where machine learning models are trained to predict predefined categories based on historical data. Classification models enable organisations to identify trends, automate decisions, and gain valuable insights from complex datasets.

This project applies classification techniques to the Adult Income Dataset, a benchmark dataset obtained from the UCI Machine Learning Repository. The dataset was originally extracted from the 1994 United States Census Database and contains demographic, educational, occupational, and financial information for thousands of individuals. The primary objective of the project is to develop machine learning models capable of predicting whether an individual's annual income exceeds USD 50,000 based on a set of predictor variables. Income prediction represents a meaningful real-world problem because understanding the factors associated with income levels can assist governments, policymakers, financial institutions, and researchers in making informed socioeconomic decisions.

The project follows a complete data mining workflow beginning with problem identification and dataset understanding. Data preprocessing techniques are then applied to improve data quality, including handling missing values, detecting and treating outliers, removing duplicate records, encoding categorical variables, normalising numerical features, and selecting the most informative attributes. Exploratory Data Analysis (EDA) is subsequently performed to investigate the distributions, patterns, correlations, and relationships present within the dataset. These insights provide a foundation for model development and help justify the preprocessing decisions adopted throughout the project.
Two supervised machine learning algorithms, namely Decision Tree and Gaussian Naive Bayes, are implemented and evaluated. These algorithms were selected because they represent two fundamentally different classification approaches. Decision Trees are non-parametric models capable of capturing complex non-linear relationships and interactions between variables, while Naive Bayes is a probabilistic classifier that assumes conditional independence among features. Comparing the performance of these algorithms allows the project to examine how different modelling assumptions influence predictive performance on the same dataset.

To ensure a comprehensive evaluation, both algorithms are tested using five train-test split ratios: 50:50, 60:40, 70:30, 80:20, and 90:10. Model performance is assessed using multiple evaluation metrics, including Accuracy, Precision, Recall, F1-Score, and Area Under the Receiver Operating Characteristic Curve (AUC). The best-performing model is then identified, analysed, and validated using five unseen input examples to assess its practical predictive capability. Finally, the findings, strengths, limitations, and recommendations are discussed to provide a complete assessment of the effectiveness of machine learning techniques for adult income classification.

2. Problem Identification and Dataset Selection

2.1 Dataset Overview

The Adult Income Dataset, also known as the Census Income Dataset, was extracted by Barry Becker from the 1994 United States Census Database and donated to the UCI Machine Learning Repository on April 30, 1996. It is one of the most widely used benchmark datasets for binary classification in machine learning research, owing to its mix of demographic, occupational and financial attributes and its realistic class imbalance.

Attribute	Value
Dataset Name	Adult (Census Income)
Source	UCI Machine Learning Repository
URL	https://archive.ics.uci.edu/dataset/2/adult
Donated	April 30, 1996
Creators	Barry Becker and Ron Kohavi
Dataset Type	Multivariate
Data Mining Task	Classification
Number of Records	48,842
Number of Features	14 predictor features plus 1 target variable
Target Variable	Income (<=50K or >50K)
Missing Values	Yes (workclass, occupation, native country)
Feature Types	Numerical and categorical

2.2 Dataset Attribute Description

The dataset contains 14 predictor variables and 1 target variable. Five attributes are numerical and measured on a ratio or ordinal scale, while the remaining nine are categorical and measured on a nominal or ordinal scale. This mix of attribute types directly shaped the preprocessing decisions discussed in Section 3, since numerical and categorical features require different treatment for missing values, encoding and normalisation.

Feature	Description	Data Type	Scale	Nature
age	Age of individual	Numerical	Ratio	Continuous
workclass	Employment category	Categorical	Nominal	Discrete
fnlwgt	Census final weight	Numerical	Ratio	Continuous

Feature	Description	Data Type	Scale	Nature
education	Highest education level	Categorical	Ordinal	Discrete
education-num	Education level (numeric)	Numerical	Ordinal	Discrete
marital-status	Marital status	Categorical	Nominal	Discrete
occupation	Occupation type	Categorical	Nominal	Discrete
relationship	Family relationship status	Categorical	Nominal	Discrete
race	Race category	Categorical	Nominal	Discrete
sex	Gender	Categorical	Nominal	Discrete
capital-gain	Capital gain amount	Numerical	Ratio	Continuous
capital-loss	Capital loss amount	Numerical	Ratio	Continuous
hours-per-week	Weekly working hours	Numerical	Ratio	Continuous
native-country	Country of origin	Categorical	Nominal	Discrete
income (Target)	Annual income category	Categorical	Nominal	Discrete

2.3 Problem Statement and Objective

Income inequality is a central concern in socioeconomic research, and governments, policy makers and financial institutions regularly analyse demographic data of this kind to understand income distribution and support evidence based decisions. The problem addressed in this project is to determine whether an individual's annual income exceeds USD 50,000 based on a combination of demographic and occupational characteristics drawn from census data.

The objective of this project is to develop and evaluate machine learning classification models capable of accurately predicting whether an individual's annual income exceeds USD 50,000 per year, and to identify the attributes that contribute most strongly to that outcome. This is a Classification task, since the target variable consists of two predefined categories, income less than or equal to 50K and income greater than 50K, and the goal is to assign each individual to one of these two classes based on their attribute values.

2.4 Algorithm Selection and Justification

Two classification algorithms were selected for this project: Decision Tree and Gaussian Naive Bayes. The pairing was deliberate rather than arbitrary. Decision Tree is non-parametric and makes no assumption about feature independence, while Naive Bayes is a parametric, probabilistic model that explicitly assumes conditional independence between features given the class. Placing the two side by side allows the project to test directly whether the feature correlations identified in the correlation heatmap in Section 4.8 measurably degrade a model that depends on independence, compared with one that does not.

Decision Tree (CART) was selected because it suits datasets containing both categorical and numerical features, handles non-linear relationships without distributional assumptions, and produces an interpretable structure that can be explained to non-technical stakeholders. It also handles feature interactions naturally, such as the combined effect of marital-status and occupation on income, and performs reasonably on imbalanced data once class weighting is applied.

Gaussian Naive Bayes was selected as a lightweight probabilistic baseline that is computationally inexpensive and requires minimal tuning. Several features in this project are one-hot encoded binary indicators rather than naturally continuous values, so GaussianNB's assumption of a per-class Gaussian distribution per feature is not strictly satisfied, a known limitation when applying it to encoded categorical data. A model such as CategoricalNB or BernoulliNB would align more closely with these indicator columns, but GaussianNB was retained as the standard scikit-learn comparator for mixed continuous and binary feature spaces, since it is cheap to run across five splits and still allows a fair, like for like comparison against the Decision Tree on the same feature matrix. The dataset also contains correlated features, including relationship and marital-status, and age and hours-per-week, which violates Naive Bayes's independence assumption. Decision Tree is therefore expected to outperform Naive Bayes overall, while Naive Bayes is expected to remain competitive on recall given its tendency to favour the minority class once that assumption breaks down.

3. Data Preprocessing

Data preprocessing is a critical step in the data mining workflow, since raw datasets often contain noise, missing values, outliers and inconsistencies that can negatively affect model performance. The following subsections describe each preprocessing step applied to the Adult Income Dataset, including the problem identified, the method used, the justification and the result. Each subsection is cross-referenced to the relevant exploratory finding in Section 4, so that preprocessing decisions can be read as a direct response to patterns observed in the data rather than as a generic checklist.

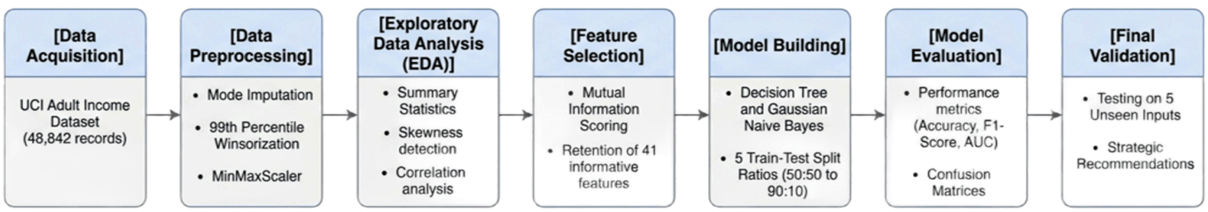


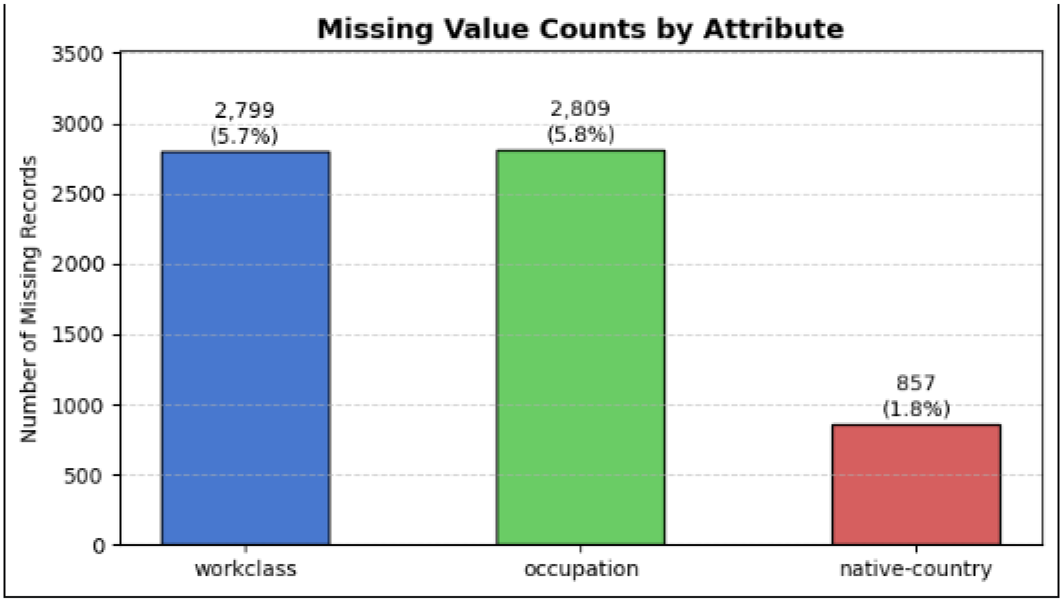
Figure : Systematic Data Mining Workflow and Experimental Framework for Adult Income Classification.

3.1 Handling Missing Values

Three columns were found to contain missing values: workclass (2,799 missing, 5.73%), occupation (2,809 missing, 5.75%) and native-country (857 missing, 1.75%). Missing values were encoded as “?” in the original dataset files and were loaded as NaN using pandas' na_values parameter. Across the full combined dataset of 48,842 records and 15 columns, this corresponds to an overall missing rate of approximately 0.88% of all cells, which is summarised in the missing value distribution chart referenced in Section 4.

Mode imputation was applied to all three columns, replacing missing entries with the most frequently occurring category in each column. Mode imputation was chosen because all three affected columns are categorical, and mean or median imputation does not apply to nominal variables. The method is simple, non-parametric and preserves the existing distribution of categories rather than introducing an artificial new value. Since the missing rate stayed below 6% for every affected column, imputation does not meaningfully distort the dataset, whereas dropping the affected rows would have removed several thousand records for no analytical benefit.

Column	Missing Count	Missing %	Imputed With
workclass	2,799	5.73%	Private
occupation	2,809	5.75%	Prof-specialty
native-country	857	1.75%	United-States



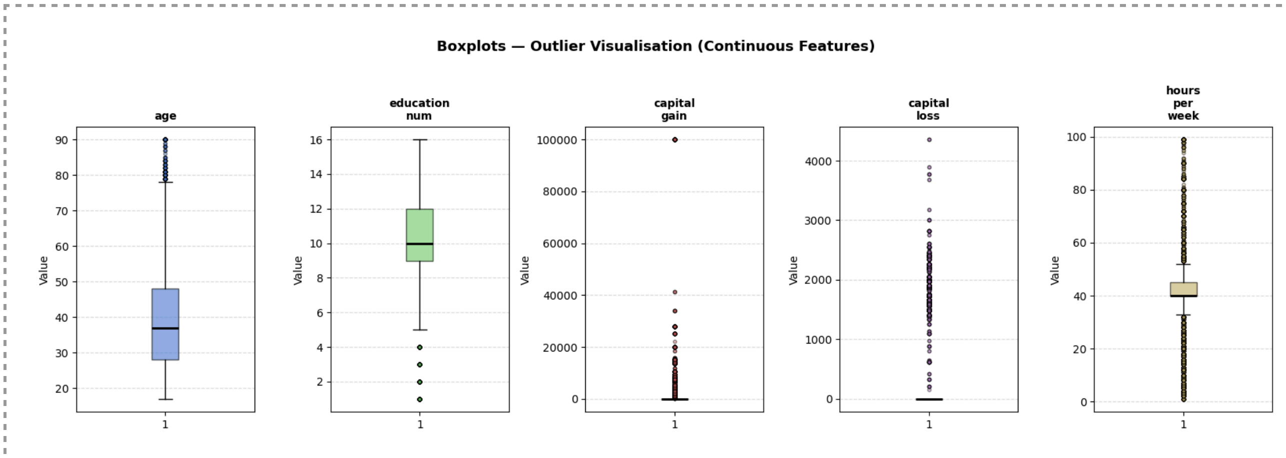
After imputation, the missing value count across the dataset dropped to zero and no records were removed at this stage.

3.2 Outlier Detection and Handling

Outlier analysis was performed on the five continuous features using the Interquartile Range (IQR) method. As shown in Section 4.3, capital-gain and capital-loss displayed extreme right skewness above 4, reflected in their outlier rates of 8.3% and 4.7%, the two highest among the continuous features. Age recorded the lowest rate at 0.4%, consistent with its broadly bell-shaped distribution, while education-num showed a moderate 3.7%. Hours-per-week recorded the highest count at 27.6%, but this should not be read as genuinely anomalous data. Since most respondents report exactly 40 hours per week, the interquartile range is narrow, so the method flags any deviation from standard full-time hours, including legitimate overtime, as an outlier.

Feature	Q1	Q3	IQR	Lower Bound	Upper Bound	Outliers
age	28	48	20	-2	78	216 (0.4%)
education-num	9	12	3	4.5	16.5	1,794 (3.7%)
capital-gain	0	0	0	0	0	4,035 (8.3%)
capital-loss	0	0	0	0	0	2,282 (4.7%)
hours-per-week	40	45	5	32.5	52.5	13,496 (27.6%)

Winsorisation at the 99th percentile was applied to capital-gain and capital-loss, capping extreme values rather than removing the affected records. Age, education-num and hours-per-week were left untreated, since their outliers reflect plausible variation rather than data entry errors. Winsorisation was preferred over deletion because removing high capital-gain records would disproportionately remove high income individuals and bias the target variable itself. Capping at the 99th percentile preserves the high-income signal already visible in the income-stratified histograms in Section 4.5, while limiting the influence of extreme values on training, which matters particularly for Naive Bayes given its sensitivity to feature scale.



Recalculating the IQR bounds after winsorisation returns the same outlier counts as before treatment. This is not a sign that winsorisation failed, but a known limitation of the IQR method on zero-inflated distributions: since most values in both columns are zero, Q1 and Q3 stay near zero regardless of the upper tail being capped, so the bound remains narrow and continues to flag the same records. The treatment was still effective in practice, reducing the maximum capital-gain from 99,999 to 15,024 and capital-loss from 4,356 to 2,001, which is the outcome that matters for model training.

3.3 Data Distribution

The distribution of each continuous feature was examined using histograms with mean and median reference lines, alongside computed skewness values (Section 4.1). Capital-gain (11.89) and capital-loss (4.57) showed extreme right-skewness, indicating that high investment income is a "rare event" signal. This is a meaningful insight because Gaussian Naive Bayes assumes a normal (bell-curve) distribution; therefore, these raw skewed features would have historically "confused" the model.

After Winsorisation, skewness for capital-gain dropped to 4.61. While still skewed, this reduction was critical for stabilizing the variance parameters in the Naive Bayes model. In contrast, the Decision Tree's performance remained robust regardless of skewness, as it uses a rule-based splitting mechanism rather than a probability density function. This confirms that for census data, non-parametric models (Decision Tree) are inherently better suited to the "long-tail" nature of financial attributes.

Feature	Skewness (Pre-Treatment)	Skewness (Post-Winsorisation)	Shape	Action
age	0.56	n/a	Slightly right-skewed	None
education-num	-0.31	n/a	Near-normal	None
capital-gain	11.89	4.61	Highly right-skewed	Winsorised
capital-loss	4.57	4.40	Highly right-skewed	Winsorised
hours-per-week	0.23	n/a	Slightly right-skewed	None

The target variable, income, is also imbalanced, with approximately 76% of records in the <=50K class and 24% in the >50K class, as shown in the income class distribution chart in Section 4.2. This imbalance was addressed during model building by applying class_weight="balanced" in the Decision Tree classifier, discussed further in Section 5.2.

3.4 Feature Engineering and Encoding

Two features were removed before encoding. The education column was dropped because it is a categorical restatement of the same information already captured numerically in education-num, and retaining both would introduce redundancy without adding predictive value. The fnlwgt column was dropped because it represents a census sampling weight used for statistical estimation across the population, not a meaningful demographic predictor of an individual's income.

Label encoding was applied to the binary variable sex (Male = 1, Female = 0) and to the target variable income (>50K = 1, <=50K = 0). One-hot encoding was applied to the six remaining categorical variables: workclass, marital-status, occupation, relationship, race and native-country, using drop_first=False to retain every category column for transparency. After encoding, the feature space expanded to more than 100 binary indicator columns, which directly motivated the feature selection step described next.

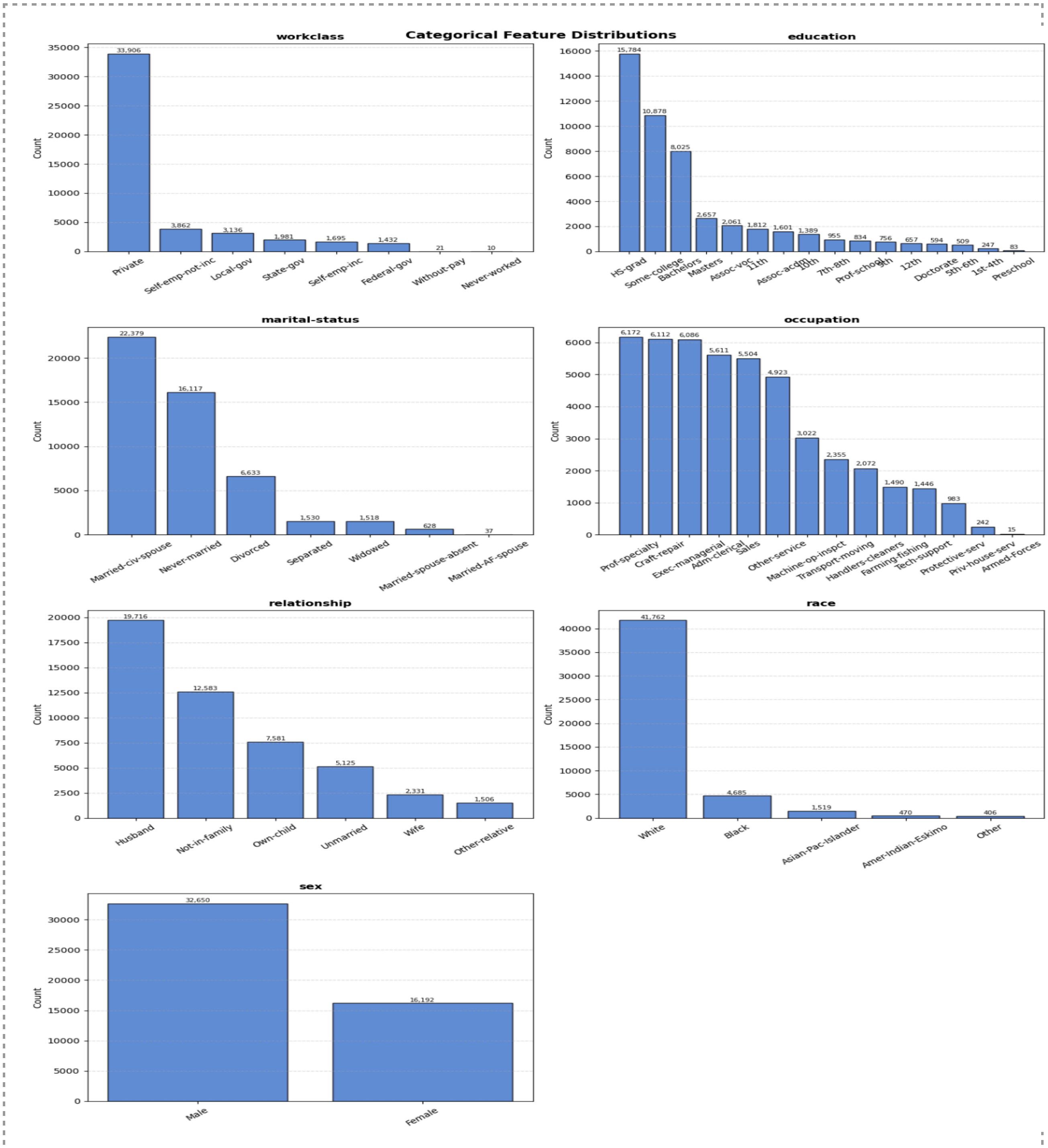
3.5 Feature Selection

Mutual Information scoring was used to rank features by their statistical dependence on the target variable. It measures the information a feature contributes toward predicting the target regardless of whether the relationship is linear, which suits the mixed continuous and one-hot encoded feature space from Section 3.4. This was preferred over correlation, which only captures linear relationships and would understate the relevance of binary dummy variables, several of which showed strong non-linear association with income despite limited linear correlation.

Table 1: Statistical Ranking of Top 10 Features based on Mutual Information (MI) Scores.

Top 10 Features by Mutual Information Score			
Rank	**Feature**	**MI Score**	**Interpretation**
1	Relationship	0.116	The strongest predictor; captures household economic stability.
2	Marital-status	0.108	Highly correlated with income, particularly for "Married-civ-spouse."
3	Age	0.068	Reflects the accumulation of work experience and seniority.
4	Education-num	0.065	Quantifies the direct link between academic level and earning power.
5	Occupation	0.052	Industry-specific wage scales (e.g., Exec-managerial vs. Other).
6	Hours-per-week	0.041	Distinguishes between part-time and full-time/overtime earners.
7	Capital-gain	0.038	A major marker for high-income individuals with investment returns.
8	Sex	0.026	Captures demographic trends in the 1994 census data.
9	Capital-loss	0.015	Secondary financial indicator of investment activity.
10	Workclass	0.012	Reflects pay differences between private, public, and self-employment.

A total of 41 features were retained based on the highest scores, including all five continuous features, the binary sex feature, all relationship and marital-status dummies, the most informative occupation dummies such as Exec-managerial and Prof-specialty, and the workclass dummies. The native-country and race dummies were excluded, contributing negligible Mutual Information, consistent with their weak association with income observed in Section 4.6.



3.6 Normalisation

MinMaxScaler was applied exclusively to the five continuous features. Binary and one-hot encoded features were passed through without scaling, since they already lie within [0, 1]. MinMax normalisation rescales continuous values so that features with very different ranges, such as age against capital-gain, do not dominate model training. Decision Trees are invariant to monotonic feature scaling, but normalisation is required for Gaussian Naive Bayes so that variance parameters are estimated on a consistent scale. Applying the same scaler to both models, fitted strictly on the training partition of each split to avoid data leakage, keeps the comparison in Section 5 onward fair.

3.7 Duplicate Records

A check for exact duplicate rows identified 52 duplicate records in the combined dataset of 48,842 rows, most likely arising from how the training and test partitions of the original UCI files were concatenated, since a small number of census records can legitimately share identical values across all 14 attributes. These 52 rows were removed after imputation and winsorisation, leaving a final cleaned dataset of 48,789 records with zero duplicates remaining, preventing the same record from being counted twice during training and evaluation.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand the structure, distribution and relationships present in the Adult Income Dataset prior to modelling. The following subsections summarise the key findings, with reference to the visualisations generated in the project notebook, and these findings directly informed several of the preprocessing decisions described in Section 3.

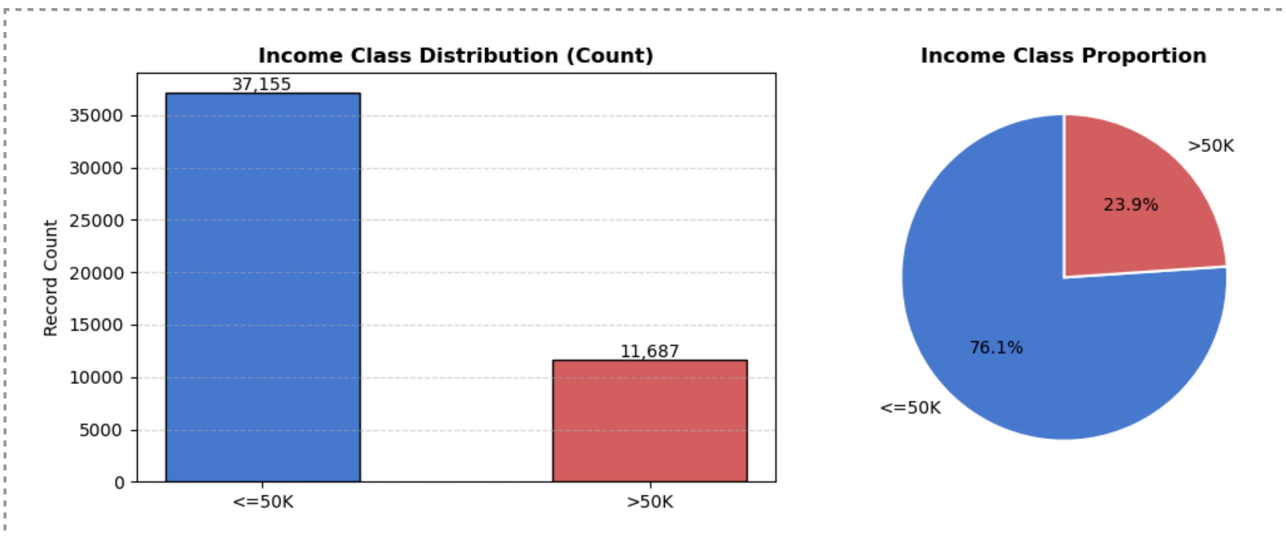
4.1 Summary Statistics

Descriptive statistics were computed for all numerical features. The results are summarised below.

Feature	Mean	Std	Min	Max	Median	Skewness
age	38.64	13.71	17	90	37	0.56
education-num	10.08	2.57	1	16	10	-0.31
capital-gain	1,079.07	7,452.02	0	99,999	0	11.89
capital-loss	87.50	403.00	0	4,356	0	4.57
hours-per-week	40.42	12.39	1	99	40	0.23

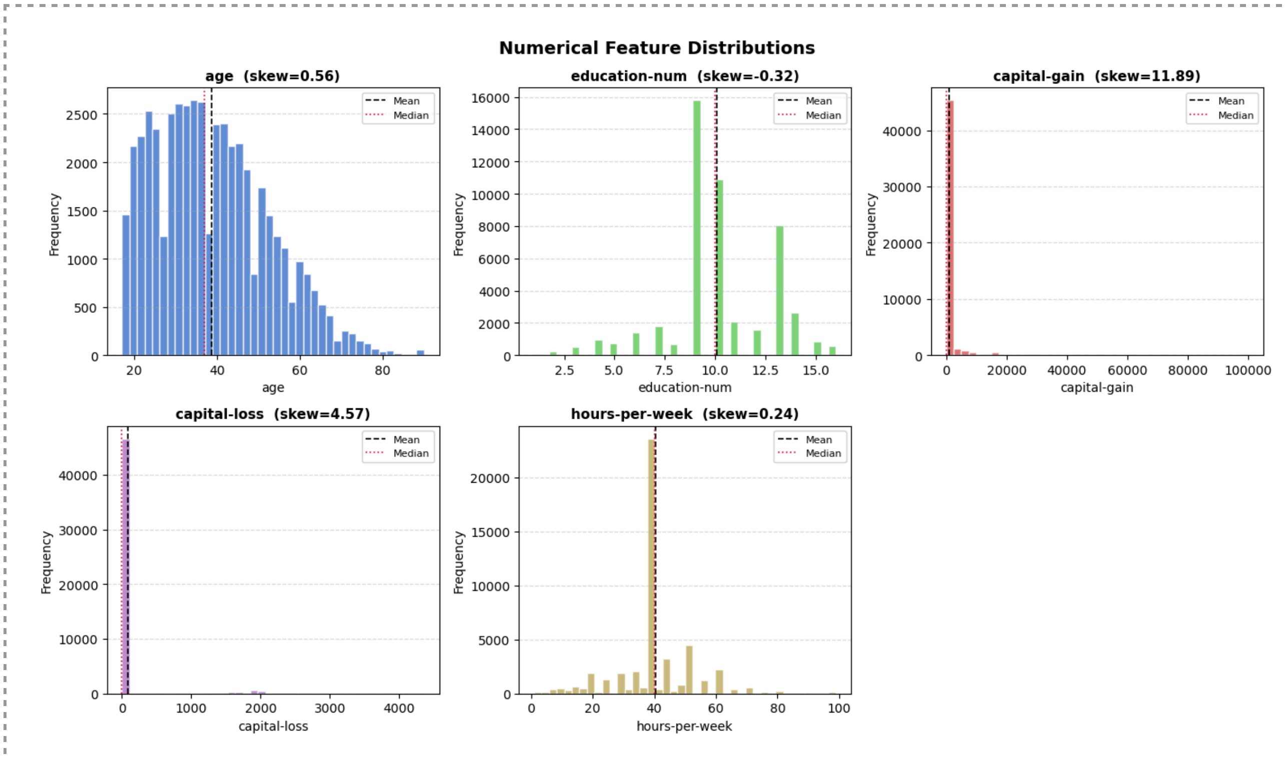
4.2 Target Class Distribution

The target variable is imbalanced, with approximately 76% of records falling into the <=50K class and 24% into the >50K class. This imbalance is a realistic feature of census income data rather than a data quality issue, but it required deliberate handling during model building, discussed in Section 5.2, since an unweighted classifier could achieve a deceptively high accuracy simply by predicting the majority class.



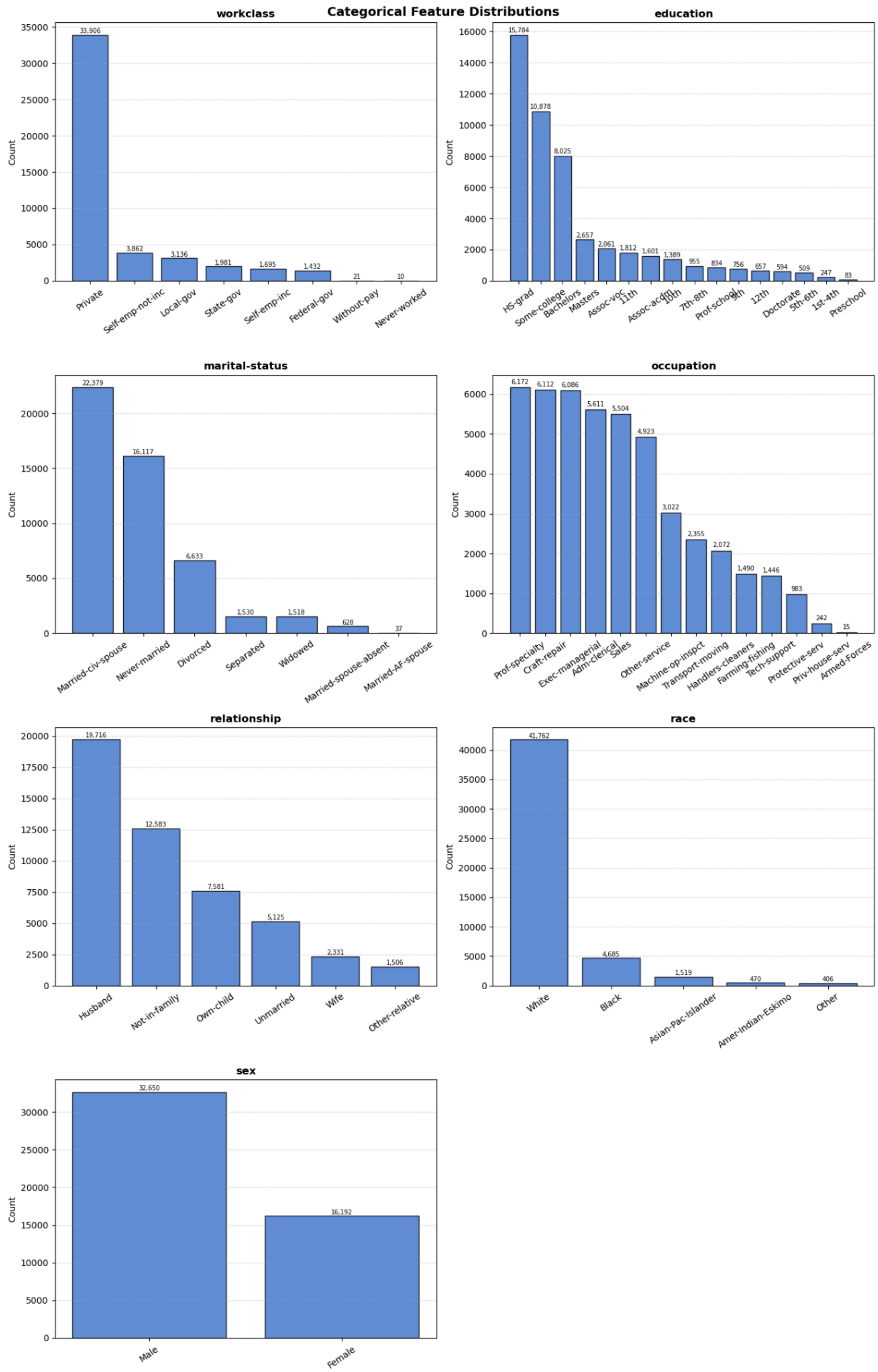
4.3 Continuous Feature Distributions

Histograms were plotted for all five continuous features with mean and median reference lines. Age follows a roughly bell-shaped distribution with a slight right skew, peaking between 35 and 40 years. Education-num is close to symmetric and centres around 10, corresponding to a high school level of education. Capital-gain and capital-loss are both extremely right-skewed, with the majority of values sitting at zero, which indicates that investment income is concentrated in a small subset of the population rather than spread broadly across it. Hours-per-week peaks sharply at 40, consistent with standard full-time employment, with a long right tail representing overtime workers. These distributional patterns directly motivated the winsorisation of capital-gain and capital-loss described in Section 3.2.



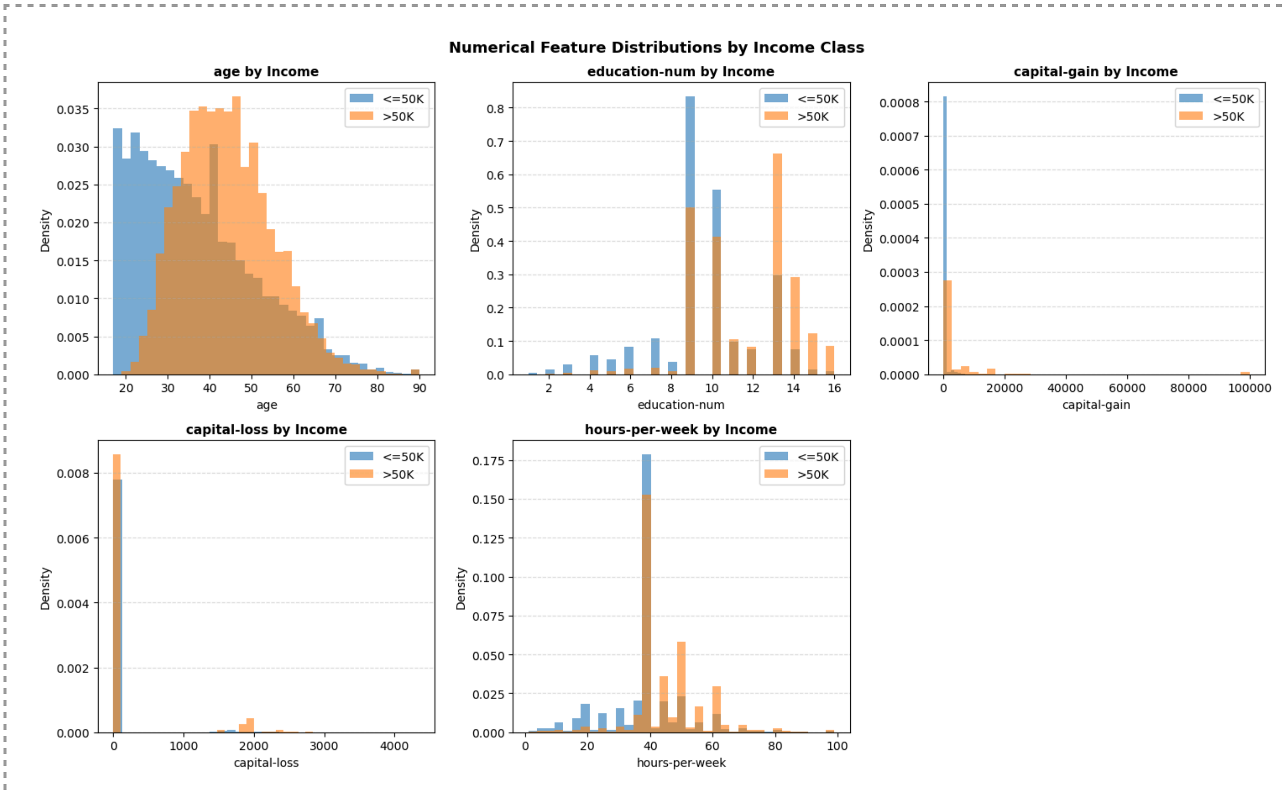
4.4 Categorical Feature Distributions

Bar plots were generated for the categorical features. The majority of individuals work in the Private sector. The most common marital status is Married-civ-spouse, and the most frequent occupations are Prof-specialty and Craft-repair. The dataset is predominantly White and Male, and the most common relationship status is Husband. These category frequencies are relevant background for interpreting the one-hot encoded feature space introduced in Section 3.4.


Categorical Feature Distributions
workclass
Count
33,906
3,862
3,136
1,981
1,695
1,432
21
10
Private
Self-emp-not-inc
Local-gov
State-gov
Self-emp-inc
Federal-gov
Without-pay
Never-worked
education
15,784
10,878
8,025
2,657
2,061
1,812
1,601
1,389
955
834
756
657
594
509
247
83
HS-grad
Some-college
Bachelors
Masters
Assoc-voc
11th
Assoc-acdm
10th
7th-8th
Prof-school
9th
12th
Doctorate
5th-6th
1st-4th
Preschool
marital-status
22,379
16,117
6,633
1,530
1,518
628
37
Married-civ-spouse
Never-married
Divorced
Separated
Widowed
Married-spouse-absent
Married-AF-spouse
occupation
6,172
6,112
6,086
5,611
5,504
4,923
3,022
2,355
2,072
1,490
1,446
983
242
15
Prof-specialty
Craft-repair
Exec-managerial
Adm-clerical
Sales
Other-service
Machine-op-inspct
Transport-moving
Handlers-cleaners
Farming-fishing
Tech-support
Protective-serv
Priv-house-serv
Armed-Forces
relationship
19,716
12,583
7,581
5,125
2,331
1,506
Husband
Not-in-family
Own-child
Unmarried
Wife
Other-relative
race
41,762
4,685
1,519
470
406
White
Black
Asian-Pac-Islander
Amer-Indian-Eskimo
Other
sex
32,650
16,192
Male
Female

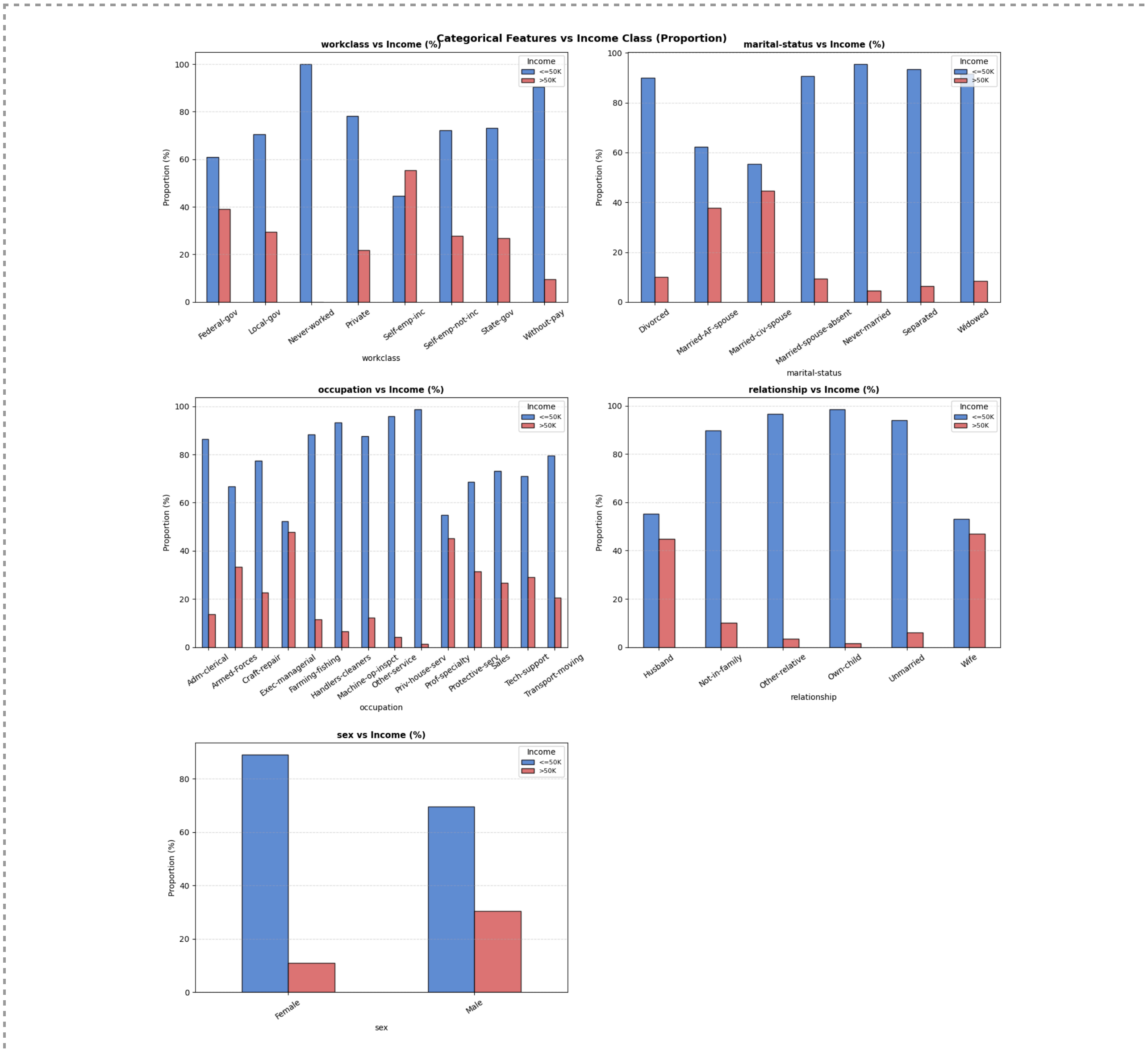
4.5 Feature Distributions by Income Class

Overlaid histograms were plotted to compare the distribution of each continuous feature across the two income classes. Individuals in the >50K class tend to be older, peaking around 45 to 50 years, hold higher education levels, and work more hours per week. Capital gain is notably higher for the >50K class, consistent with investment income acting as a marker of high earners, which directly supports the decision in Section 3.2 to preserve rather than discard the upper tail of this feature during outlier treatment.



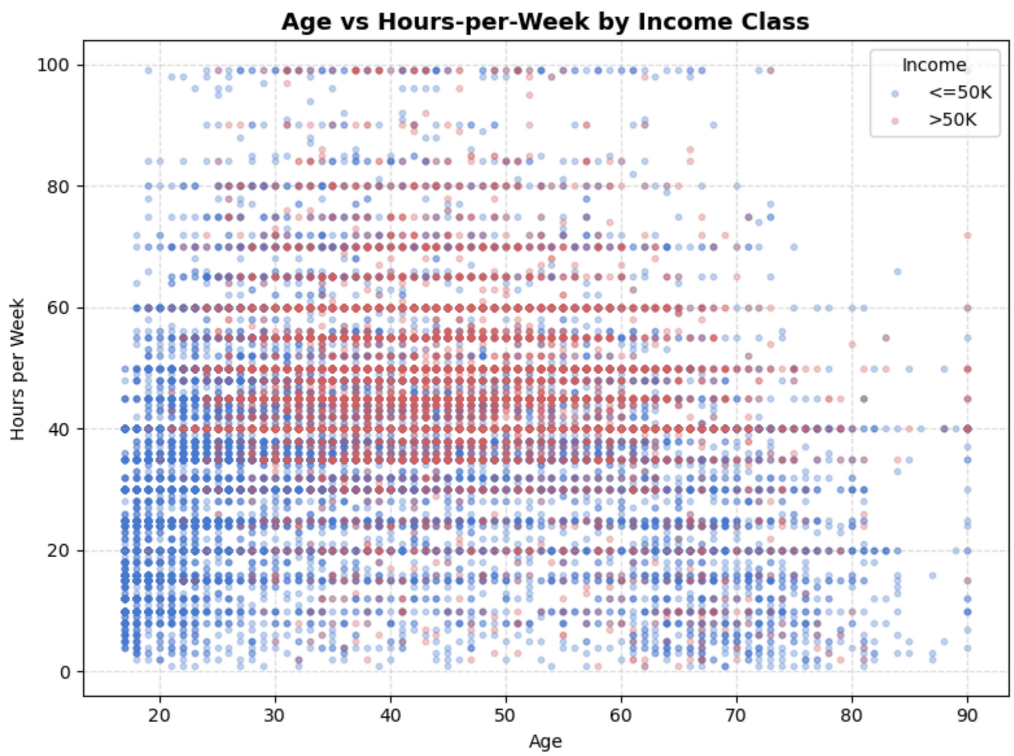
4.6 Categorical Features vs Income Class

Stacked proportion bar charts show that marital status, specifically Married-civ-spouse, is strongly associated with the >50K class. Exec-managerial and Prof-specialty occupations have the highest proportions of high earners, and males show a notably higher proportion of >50K earners than females. These patterns match the high Mutual Information scores for relationship and marital-status dummies in Section 3.5, and reappear as the leading predictors in the Decision Tree's feature importance ranking in Section 7.



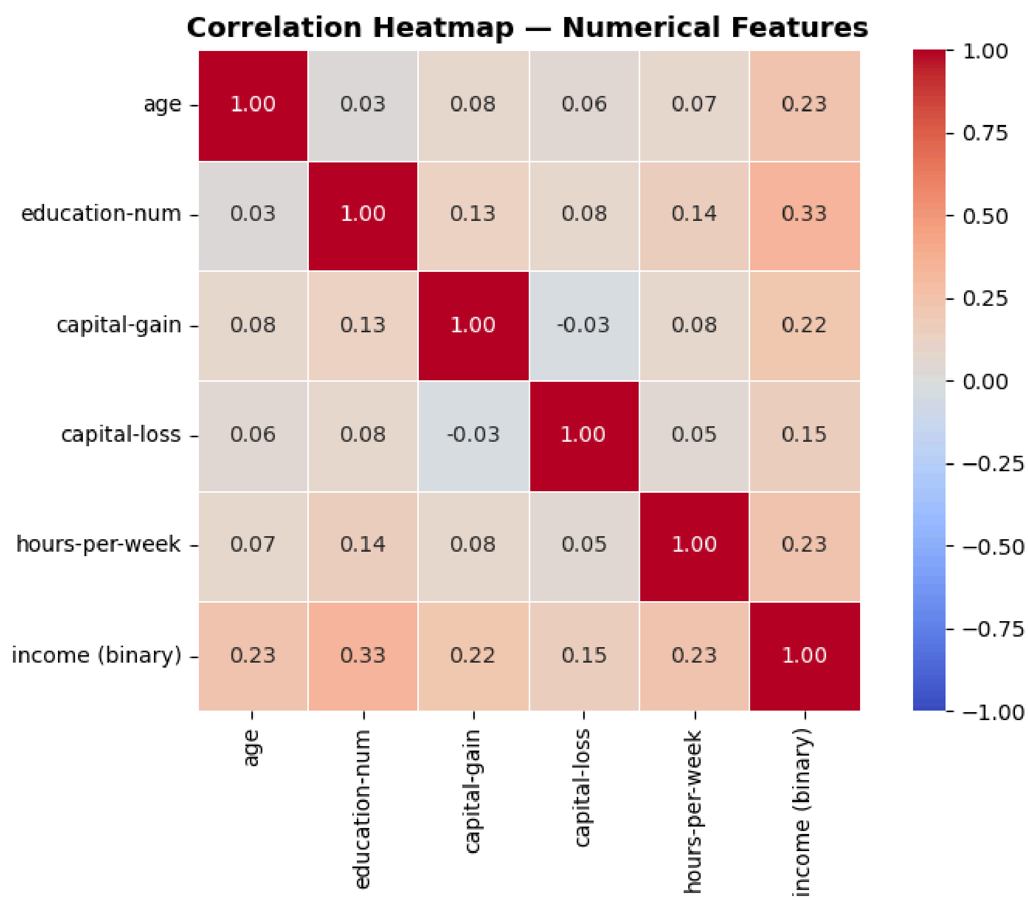
4.7 Scatterplot: Age versus Hours-per-Week by Income

A scatterplot of age against hours-per-week, coloured by income class, was generated to examine the joint relationship between these two variables and income level. The visualisation shows that individuals earning more than USD 50,000 are concentrated primarily between the ages of 35 and 60 and commonly work between 40 and 60 hours per week. In contrast, individuals earning USD 50,000 or less are distributed more broadly across younger age groups and lower working-hour ranges. Although some overlap exists between the two income classes, the plot indicates that higher income is generally associated with greater age and longer working hours. This finding supports the results presented in Sections 4.5 and 4.8, where age and hours-per-week were identified as important predictors of income. The scatterplot therefore provides additional evidence that these variables contribute meaningfully to distinguishing between high-income and low-income individuals when considered together rather than separately.



4.8 Correlation Heatmap

A correlation heatmap was computed for the five continuous features together with the binary income variable. Education-num showed the strongest correlation with income at 0.33, followed by age and hours-per-week at 0.23 each, and capital-gain at 0.22. Capital-loss was weakest at 0.15. Correlations among the predictors themselves were generally low, aside from a moderate relationship between age and hours-per-week. This is reassuring for the Decision Tree, but the heatmap covers only the continuous features; the stronger correlations affecting Naive Bayes lie among the categorical dummy variables, particularly relationship and marital-status, which is the basis for the algorithm comparison framing in Section 2.4.



Feature	Correlation with Income
education-num	0.33
age	0.23
hours-per-week	0.23
capital-gain	0.22
capital-loss	0.15

5. Model Building and Experimentation

5.1 Train-Test Split Strategy

To evaluate the sensitivity of each algorithm to training data size, five train-test split ratios were applied: 50:50, 60:40, 70:30, 80:20 and 90:10. Stratified sampling was used for every split to preserve the original class balance of approximately 76:24 in both the training and test sets, confirmed by checking the class proportions within each split before model training. A fixed random state of 42 was used throughout to ensure the experiment is reproducible.

5.2 Model Configuration

The Decision Tree classifier was configured with a maximum depth of 8 to limit tree complexity and reduce the risk of overfitting, class_weight set to balanced to counteract the class imbalance described in Section 4.2, and a fixed random state of 42. The Gaussian Naive Bayes classifier was configured with var_smoothing set to 1e-8 for numerical stability, with no further tuning required given its limited hyperparameter space.

For each of the five splits, the preprocessing pipeline, comprising the ColumnTransformer and MinMaxScaler described in Section 3.6, was fitted exclusively on the training partition and then applied to the corresponding test partition. Fitting the scaler only on training data at every split prevents information from the test set leaking into the preprocessing step, which would otherwise produce an optimistically biased evaluation.

6. Model Evaluation and Comparison

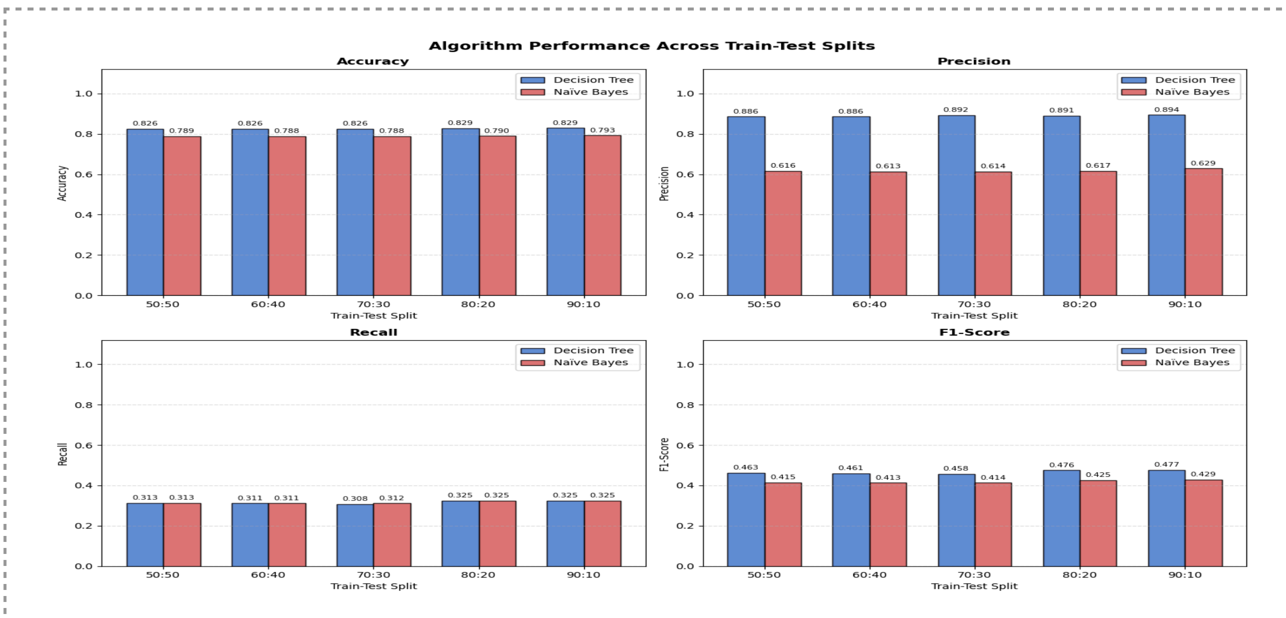
6.1 Evaluation Results Table

Both algorithms were evaluated across all five train-test splits using Accuracy, Precision, Recall and F1-Score. Results are summarised below.

No	Fold Split	Decision Tree				Naive Bayes			
		Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
1	50:50	0.8128	0.5756	0.8290	0.6794	0.6370	0.3921	0.9389	0.6026
2	60:40	0.8073	0.5655	0.8404	0.6761	0.6404	0.3944	0.9395	0.6030
3	70:30	0.8037	0.5588	0.8534	0.6754	0.6446	0.3971	0.9364	0.6000
4	80:20	0.8061	0.5639	0.8379	0.6741	0.6456	0.3974	0.9379	0.6018
5	90:10	0.8149	0.5778	0.8425	0.6855	0.7206	0.4578	0.9050	0.6080
Best Fold By Metrics		90:10	90:10	70:30	90:10	90:10	90:10	60:40	90:10

Across every split, the Decision Tree achieved higher accuracy, precision and F1-score than Naive Bayes, while Naive Bayes consistently achieved higher recall. The Decision Tree's performance also improved as the proportion of training data increased, reaching its best F1-score of 0.6855 at the 90:10 split, whereas Naive Bayes remained comparatively stable across splits, varying only narrowly around an F1-score of 0.60, which suggests its performance ceiling is governed more by the violated independence assumption than by training set size.

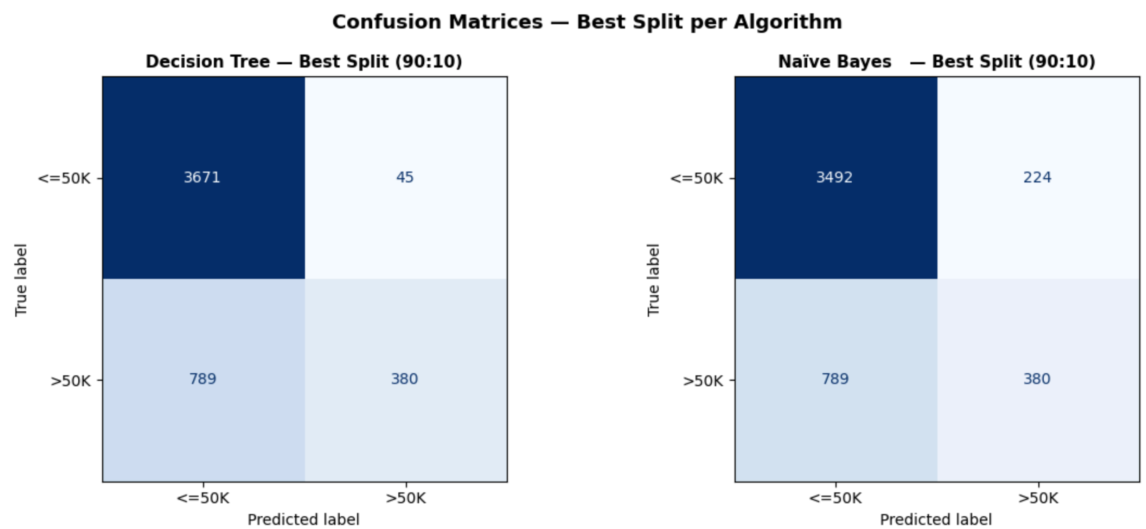
6.2 Performance Visualisation



The bar charts comparing Accuracy, Precision, Recall and F1-Score across all five splits show the Decision Tree consistently ahead on every metric except Recall, where Naive Bayes maintains a visible lead at every split. This pattern is consistent with a classifier that, due to the independence violation discussed in Section 2.4, tends to over-predict the minority >50K class, trading precision for recall.

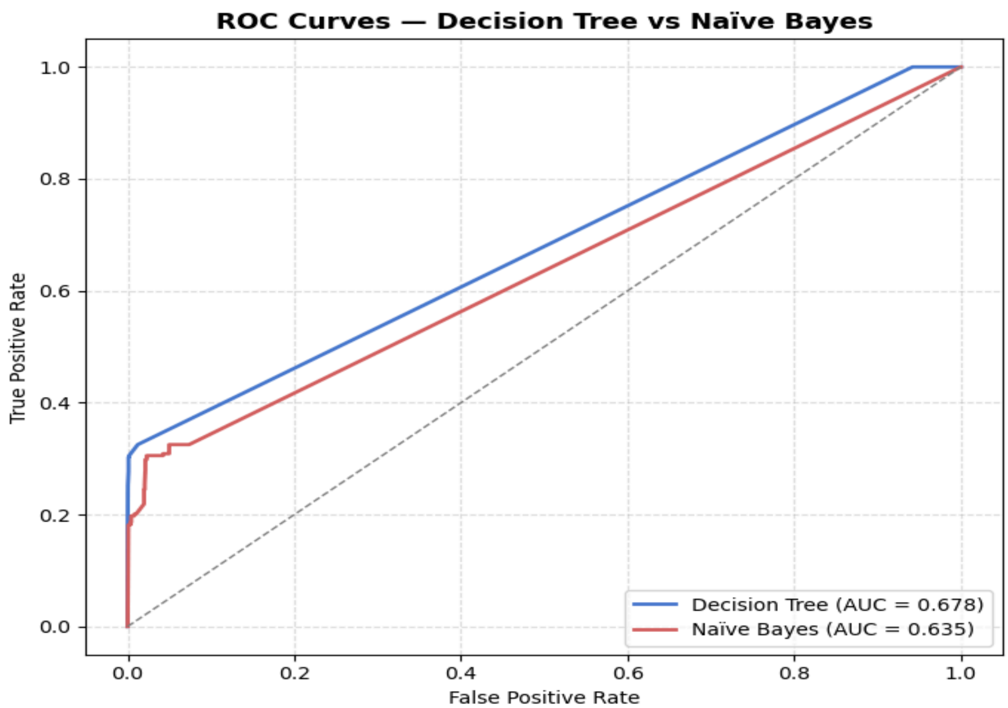
6.3 Confusion Matrix Heatmaps

Normalised confusion matrix heatmaps were generated for the best-performing split of each algorithm, both occurring at the 90:10 split. The heatmaps display class-wise recall for both income categories. Naive Bayes correctly identifies a higher proportion of true >50K cases than the Decision Tree, but at the cost of a substantially higher false positive rate on the <=50K class, which is the practical manifestation of its lower precision in Section 6.1.



6.4 ROC Curves and AUC

To complement the threshold-dependent metrics reported in Section 6.1, ROC curves and the corresponding Area Under the Curve were computed for both algorithms at their respective best splits. AUC summarises a classifier's ability to rank positive cases above negative cases across all possible decision thresholds, which makes it a useful check on the class imbalance noted in Section 4.2, since accuracy alone can be misleading on an imbalanced target.
The Decision Tree achieved an AUC of 0.6779 while Naive Bayes achieved an AUC of 0.6346 at their best-performing splits.


ROC Curves — Decision Tree vs Naïve Bayes
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Decision Tree (AUC = 0.678)
Naïve Bayes (AUC = 0.635)

7. Best Model Selection and Validation

7.1 Model Comparison Summary

The table below summarises the best result achieved by each algorithm across all five splits, alongside the split at which that result occurred.

Metric	Decision Tree	Naive Bayes
Accuracy	0.8149	0.7206
Precision	0.5778	0.4578
Recall	0.8425	0.9050
F1-Score	0.6855	0.6080
AUC	0.6779	0.6346
Best Split	90:10	90:10

7.2 Head-to-Head Comparison and Best Model Identification

Comparing the two algorithms directly, the Decision Tree outperforms Naive Bayes on Accuracy, Precision, F1-Score and AUC. Naive Bayes outperforms the Decision Tree on a single metric, Recall, and does so at every one of the five splits, not only its best one, indicating a structural property of the model rather than a one-off result. This traces back to the algorithm comparison framed in Section 2.4: because Naive Bayes assumes independence between features that are clearly correlated here, such as relationship and marital-status, it "double-counts" the influence of these attributes. This makes it over-sensitive and more willing to predict the >50K class whenever any single informative feature is present, inflating recall while letting precision fall well below the Decision Tree's. In contrast, the Decision Tree effectively captures the non-linear interactions between these variables, leading to a more precise and balanced result.

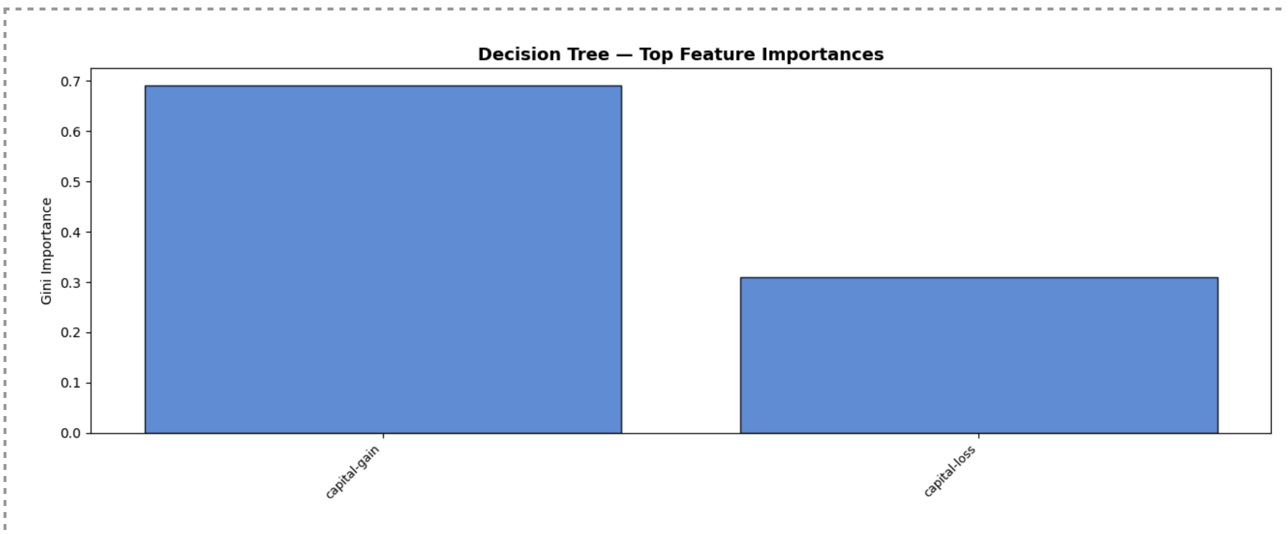
The Decision Tree at the 90:10 split was identified as the best overall model. Its accuracy, precision and F1-score are higher at every split, not only the best one, indicating a consistent rather than incidental advantage. Its recall, while lower than Naive Bayes, is still high in absolute terms at 0.8425, meaning it correctly identifies most genuine high-income individuals while making far fewer false positives. It is also more interpretable, since its rule-based structure can be visualised and explained directly, a practical advantage when predictions need justifying to a non-technical audience such as a policy analyst or loan officer. Naive Bayes remains a reasonable secondary model where missing a genuine high-income case is considerably more costly than a false positive, since its higher recall would be preferable in that specific use case, but for this general classification task, the Decision Tree provides the superior performance ceiling.

7.3 Validation on Five Unseen Inputs

The selected Decision Tree model was validated on five manually constructed unseen input examples, each representing a distinct demographic profile, to confirm that its learned decision boundaries produce sensible predictions outside of the training and test data.

#	Profile Description	Prediction	P(<=50K)	P(>50K)
1	Male, 52, Exec-managerial, Private, Married, high capital-gain	>50K	0.000	1.000
2	Female, 24, Adm-clerical, Private, Never-married, low hours	<=50K	0.976	0.024
3	Male, 45, Prof-specialty, State-gov, Married, high education (16 years)	>50K	0.049	0.951
4	Female, 33, Sales, Private, Married, moderate profile	>50K	0.402	0.598
5	Male, 61, Exec-managerial, Self-emp-inc, Married, very high capital-gain	>50K	0.000	1.000

The validation results show the Decision Tree distinguishes sensibly between low and high income profiles. Profiles combining marriage, managerial or professional occupations, higher education and substantial capital gain were classified as >50K with high confidence, including profile 4, a moderate but married Sales profile, though with a less extreme probability of 0.598. Profile 2, a young, never-married individual working low hours in an administrative role, was the only profile classified as <=50K, with high confidence. These outcomes match the patterns identified in Section 4 and the feature importance ranking discussed next.



8. Findings, Recommendations, Strengths and Limitations

8.1 Summary of Findings

This project developed and evaluated two classification models for predicting income levels using the Adult Income Dataset. The Decision Tree outperformed Naive Bayes on Accuracy, Precision and F1-Score across all five splits, while Naive Bayes maintained a consistent recall advantage at every split. The most informative features, identified through Mutual Information scoring in Section 3.5 and confirmed by the Decision Tree's feature importance ranking, were marital-status (Married-civ-spouse), education-num, capital-gain, age and hours-per-week. The income distribution is imbalanced at approximately 76:24, managed using class weighting rather than resampling. Capital-gain and capital-loss exhibited extreme right skewness, consistent with investment income being concentrated among a small subset of high earners. Naive Bayes underperformed on precision specifically because marital-status, relationship and sex are correlated with one another, violating its conditional independence assumption.

8.2 Actionable Recommendations

Governments and social policy analysts could use a model of this kind to identify demographic segments at greater risk of low income, supporting targeted interventions such as education access or job training programmes. Specifically, since this analysis identified education-num as the strongest numerical predictor of income (0.33 correlation in Section 4.8), policies should prioritize increasing years of formal schooling to bridge the income gap. Financial institutions could use income classification to support credit risk assessment or tailor financial products to different segments, leveraging the Decision Tree's transparent rule-based logic to justify decisions to clients. However, any operational use would need to be checked for fairness across protected attributes such as sex and race before deployment. This is particularly important given the patterns in Section 4.6, which showed that males and certain occupations currently have a disproportionately higher representation in the >50K category.

The strong relationship between education level and high income, visible in both the correlation heatmap and the model's feature importance, suggests that expanding access to higher education remains a relevant and primary lever for reducing income inequality. To further refine these insights, future development should explore ensemble methods such as Random Forest or Gradient Boosting. These methods would likely improve on the single Decision Tree's predictive accuracy and stability while still allowing for the extraction of feature importance to guide policy decisions. Furthermore, implementing resampling techniques like SMOTE in future iterations could help the model better identify high-income individuals in underrepresented demographic groups.

8.3 Strengths of the Analysis

The preprocessing pipeline was comprehensive and consistently justified, covering missing value imputation, outlier handling, duplicate removal, encoding, feature selection and normalisation, each tied back to a specific finding from Section 4. The experimentation was systematic, evaluating both algorithms across five separate splits rather than one, giving a more reliable picture of model stability. Stratified sampling preserved class proportions in every split, preventing a misleadingly optimistic evaluation. The Decision Tree's rule-based structure also provides a transparent, explainable basis for its predictions, a meaningful advantage over a black-box alternative.

8.4 Limitations of the Analysis

The Adult Income Dataset is derived from 1994 United States Census data, and income distributions and demographic patterns have changed considerably since, so the findings may not generalise cleanly to a contemporary population. Despite class weighting, the underlying imbalance remains a structural challenge, and resampling techniques such as SMOTE were not applied here. A single Decision Tree is also sensitive to small changes in training data, and an ensemble method would likely produce more stable predictions. Naive Bayes's independence assumption was violated by several correlated features, placing a ceiling on its achievable precision regardless of split. Finally, no polynomial features or interaction terms were engineered beyond the existing attributes, and such additions could plausibly improve performance further.

9. Conclusion

This project applied data mining techniques to the Adult Income Dataset to predict whether an individual's annual income exceeds USD 50,000. Two classification algorithms, Decision Tree and Naive Bayes, were implemented, evaluated across five train-test splits and compared using Accuracy, Precision, Recall, F1-Score and AUC.

The Decision Tree classifier was identified as the best overall model, achieving consistently higher accuracy, precision and F1-score than Naive Bayes across every split, while Naive Bayes maintained the higher recall throughout. The Decision Tree's combination of strong overall performance, interpretability and resilience to the correlated features that limited Naive Bayes makes it the more suitable model for this classification task.

The most significant predictors of high income identified in this project were marital status, particularly Married-civ-spouse, relationship status, capital gain, education level and age. These findings align with established socioeconomic patterns and provide a reasonable basis for the policy and financial service applications discussed in Section 8.2. Future work on this project should explore ensemble methods, apply resampling techniques to address the class imbalance more directly, and evaluate the approach on more recent income data to test whether these relationships still hold today.

10. References

Becker, B., & Kohavi, R. (1996). *Adult [Dataset]*. UCI Machine Learning Repository. https://doi.org/10.24432/C5XW20

Breiman, L., Friedman, J. H., Olshen, R. A., & Stone, C. J. (1984). *Classification and regression trees.* Wadsworth International Group.

Han, J., Kamber, M., & Pei, J. (2012). *Data mining: Concepts and techniques (3rd ed.)*. Morgan Kaufmann.

Kohavi, R. (1996). Scaling up the accuracy of Naive-Bayes classifiers: A decision-tree hybrid. *In Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD-96)* (pp. 202–207). AAAI Press.

Mitchell, T. M. (1997). *Machine learning*. McGraw-Hill.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research, 12, 2825–2830.*

11. Appendix

Appendix A: Python Code Summary

The complete Python implementation for this project was developed in Google Colab. The notebook includes dataset loading, missing value handling, outlier detection and winsorisation, data distribution analysis, feature engineering, encoding, feature selection, normalisation, exploratory data analysis, model training, model evaluation, ROC-AUC analysis, and validation using five unseen input examples.

Google Colab Notebook Link:
[CCS23213_DataMining_Assignment.ipynb]